



Impacts of national governments on the forest cover loss in Paraguayan Chaco between 1999 and 2021

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ABSTRACT

Changes in land use can have profound impact on the Earth's. Unsustainable land use, induced by urban sprawl, along with agricultural and environmental policies, not only significant impacts climate but also lead to deforestation and environmental degradation. A fundamental tool for tracking forest cover loss is spatial monitoring. Spatial data analysis serves as an instrument to generate public and land management policies, and also to combat illegalities in the forestry sector (e.g., illegal deforestation). The present study provides the first analysis of forest cover loss in the Paraguayan Chaco in relation to environmental and forestry legislation applied during each of the six periods of national government between the years 1999 and 2021. Using an annual database from 1999 to 2021, the changes from forest to non-forest cover in each government period were determined. The methodology used to determine the connection between forest cover losses in the Paraguayan Chaco and the environmental and forestry legislation applied during each government period consisted of a qualitative analysis of the regulations, an overview of the losses that occurred during each mandate and a comparison with annual trends of forest loss before and after each regulation came into effect. In total, 32.3% (5,748,169 ha) of the forest cover existing in 1999 had been lost by 2021. Forest loss was significantly higher (27.5%) during the government period (2008–2012). Meanwhile, the lowest rate (9.5%) was identified during the period (1999–2003). In evaluating environmental and forestry legislation for each of those periods, legal gaps that offered the possibility of multiple interpretations were found. Our results showed a relationship between poor compliance with the legislation and forest cover losses; emphasizing the need for further analysis of other variables that in turn influence the dynamics of forest loss, such as financial investments, forest fires, and leakage due to stricter environmental regulations in neighboring countries.

1. Introduction

Forests represent a critical natural resource for the sustenance of life and for the maintenance and conservation of biodiversity. They provide diverse ecosystem services, fulfill the valuable function of being carbon sinks that contribute to climate balance, and have great cultural and economic value for many countries (Köhl et al., 2015; Verijdt, 2013). Despite this, forests are under pressure from different development sectors and their economic activities (Zanetti et al., 2017), resulting in

significant forest loss (change in land use of land with forest cover during an observation period (FAO, 2020a).

Deforestation (defined as the sum of all forest losses due to the transformation of forests to other land types by natural or anthropogenic causes (FAO, 2020b)) continue to advance at a worrying pace with an annual global rate of approximately 10 million hectares between 2015 and 2020. Since the 1990s around 420 million hectares of forest have been lost globally contributing to biodiversity loss (FAO and PNUMA, 2020). According to Global Forest Watch (GFW, 2021), the main causes

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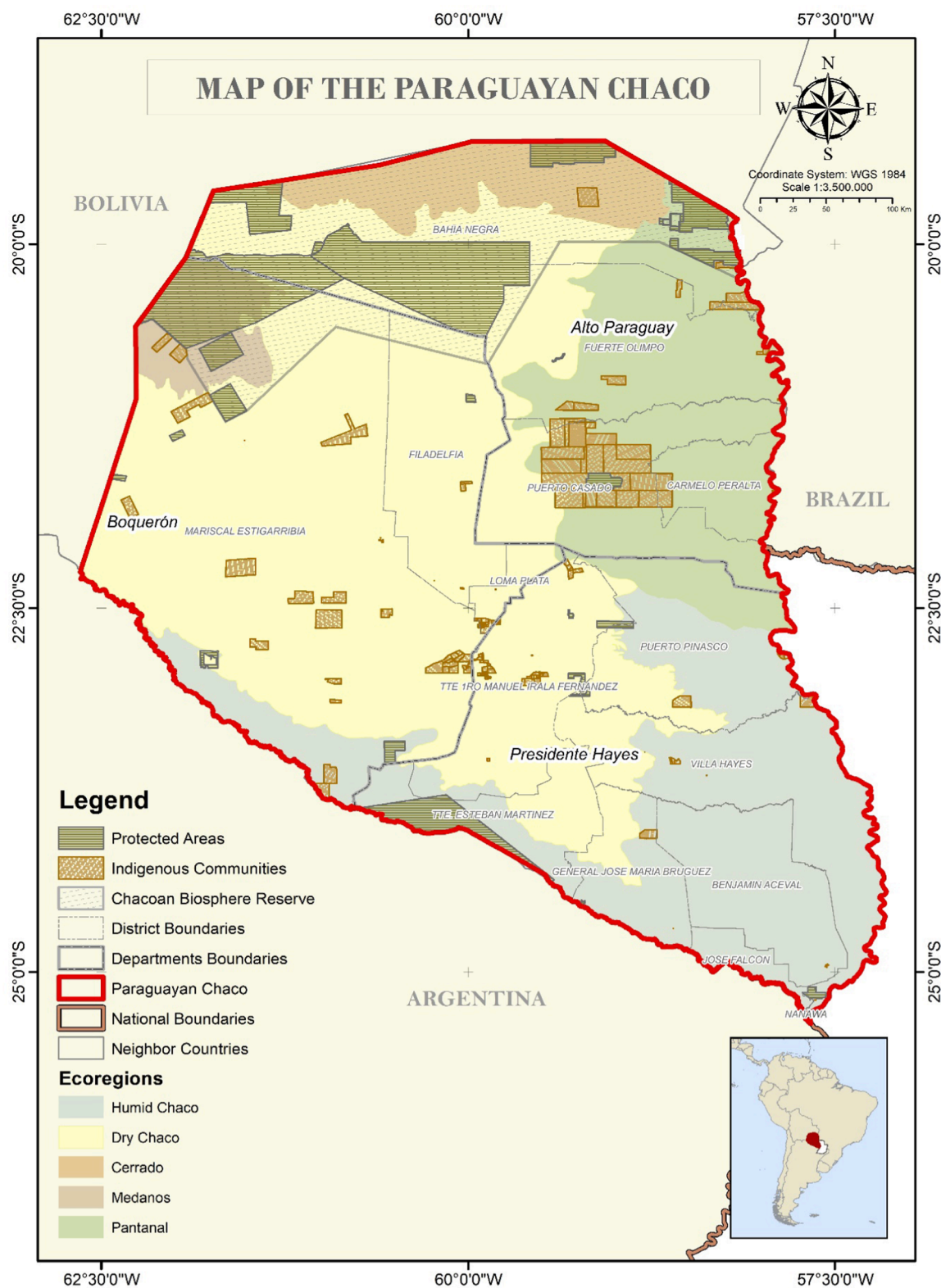
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Data Sources: Chacoan Biosphere Reserve, Protected Areas and Ecoregions: (MADES, 2021); Indigenous Communities (Gil, EA/Da Ponte, E/Insrán, KP/González, Lr, 2020); Paraguayan Chaco, Departments, District and National Boundaries (INE, 2012); South American Boundaries (<http://www.natureearthdata.com/>); Designed by Salinas Romero, S.; Casaccia Ibarrola, C.; Da Ponte, E.; García Calabrese, M. Date: December, 2021.

Fig. 1. Location Map of Study Area.

of tree cover loss in recent years in Latin America have been forestry, forest fires, agriculture, urbanization, and raw material production; the latter accounts for 27% of global forest loss (Curtis et al., 2018). This loss is also attributed to the failure of forest policies due to institutional

weaknesses and short-term political motivations that undervalue forest areas compared to other land uses, leading to negative economic and environmental consequences (Keipi, 2000).

During the first decade of the 21st century, 92% of deforestation in

Latin America occurred in South America, mainly in Argentina, Brazil, Bolivia, and Paraguay (Aide et al., 2013). Since 2000, South America has been characterized by rapid agricultural expansion as a result of the increasing global demand for soybeans and cattle, leading to the clearing of forest and savanna vegetation in the Amazon and Gran Chaco biomes (Aide et al., 2013; le Polain de Waroux et al., 2019).

The Gran Chaco constitutes the largest forests in Latin America after the Amazon and represents the second-largest biome in South America (Veit and Sarsfield, 2017). It covers parts of Argentina, Bolivia, Paraguay, and Brazil (Caldas et al., 2013; Cartes et al., 2015; Verijdt, 2013). The Gran Chaco is characterized by its important biological, cultural, and social diversity, but the richness of its territories, inhabited by different indigenous people as well as farmers, is vulnerable to deforestation caused by the advanced of the agricultural frontier (Alonso et al., 2020). This biome's dry forests are affected by some of the highest rates of deforestation worldwide (EarthSight, 2020; GFW, 2019; Vallejos et al., 2014), presenting a decrease of 20.2% (9.5 million ha) of its total forest cover between 2000 and 2019 (de la Sancha et al., 2021). Western Paraguay is the most affected region and has the highest rate of deforestation and forest cover loss (Baumann et al., 2017; de la Sancha et al., 2021; Graesser et al., 2015; Vallejos et al., 2014).

Several studies on deforestation in the Paraguayan Chaco, the country's western region, show that deforestation is clearly increasing throughout the territory (Guyra Paraguay Association, 2013; Baumann et al., 2017; Cardozo and Machado, 2019; Da Ponte et al., 2021; Mereles and Rodas, 2014), presenting forest cover losses three times greater than in the eastern region of Paraguay (GFW, 2020). This rate of deforestation is resulting in the disappearance of the landscape where the last isolated indigenous peoples of the Chaco, the Totobiegosode Ayoreos, take refuge, thus affecting the original inhabitants and the local fauna and flora (EarthSight, 2020; Ortega, 2013).

The main economic activity in the Paraguayan Chaco is agriculture, which is highly profitable for investors due to low land prices, low taxes, and low production costs (le Polain de Waroux et al., 2016; Verijdt, 2013). With livestock, agricultural, forestry, and conservation policies inefficiently organized in the country, the plurality of national policies on land use are based on the promotion of livestock and agricultural activities at the industrial level focused on exports, leading to the conversion of forest lands into pastures and crops (Kernan et al., 2010).

It must be understood that the existence of environmental policies, laws and institutions do not work without consolidated environmental and forest governance and support, equally at all levels, which means that a more in-depth analysis of specific aspects of environmental governance that guide the development of public policies is required (Wehkamp et al., 2018). Thus, knowledge, understanding and involvement of society in managing natural resources is the first step to ensuring long-term sustainability (Cobo, 2019).

Assessing forest cover by monitoring the inventory and conservation status provides key information for the design, implementation, or support of environmental policies for the protection, conservation and sustainable management of forest resources, food security and cultural aspects, among others; this, in turn, improves land use planning through sound investment and sectoral development decisions (Lwin et al., 2019; MacDicken, 2015; Romijn et al., 2015; Zanetti et al., 2017).

For this purpose, an indispensable tool is the explicit spatial description of the dynamics of transformed areas (Vallejos et al., 2014). There has been great interest in multitemporal analyses that aim to understand the spatial dynamics of forests and transformed areas around the world, including Latin America as a whole (e.g. Aide et al., 2013; Graesser et al., 2015; FAO, 2020c; Hansen et al., 2013; Keenan et al., 2015; Mahato et al., 2021; Tomas et al., 2017; Xie and Niculescu, 2021), at the Gran Chaco level (Baumann et al., 2016; de la Sancha et al., 2021; Vallejos et al., 2014), at the national level (Cabral Antúñez, 2019; Da Ponte et al., 2015; Huang et al., 2009) and within the Paraguayan Chaco (Baumann et al., 2017; Caldas et al., 2013; Mereles and Rodas, 2014; Da Ponte et al., 2021).

In recent years, attention has focused on environmental problems, their impacts and their management through policies (Cubbage et al., 2007; Ejarque, 2017). It has been shown that policies adopted by countries can have an impact on global forest loss expected by 2030 (d'Annunzio et al., 2015). There are works that focus on public policies, forest management and deforestation in different parts of the world (Dev Roy et al., 2012; Gillis, 1988; Larson and Bromely, 1991; McDermott et al., 2010), especially in Latin America (Arima et al., 2014; Assunção et al., 2012; McGinley and Finegan, 2003; Robalino et al., 2015; Zanetti et al., 2017), including comparisons between environmental governance and legislation in various countries (Lambin et al., 2014; Mansourian et al., 2014; McGinley et al., 2012; Tacconi et al., 2019), environmental regulations analyses in soybean and beef production and public policies and forest management in the region directly related to forests and their conservation (Gasparri and le Polain de Waroux, 2014; le Polain de Waroux et al., 2016; le Polain de Waroux et al., 2019) and studies on environmental regulations at local and regional levels (World Bank Group, 2018; Kernan et al., 2010; Kronawetter and Rolón, 1993; Veit and Sarsfield, 2017; World Bank, 2020).

For the most part these studies involve factors such as population dynamics, the budget allocated to environmental and forestry factors, land use plan implementation, and forest classification, but none systematically assesses the influence of environmental and forestry legislation on forest cover loss over long periods of time and grouped by periods of government. Within this context, this research seeks to analyze the impact of local policies on the loss of forest cover in the Paraguayan Chaco, combining the analysis of state laws and remote sensing data for 1999–2021. To this end, the following objectives were defined:

- Determine the loss of forest cover during each government period from 1999 to 2021.
- Study the environmental and forestry regulations in force in the Paraguayan Chaco during the different periods of government.
- Analyze the relationship between changes in forest cover and the environmental legislation applied in each case study.

2. Materials and methods

2.1. Study area

The western region of Paraguay, known as the Paraguayan Chaco, is a large plain of approximately 240,888 km² that is part of the Great American Chaco (Fig. 1). The Paraguayan Chaco is administratively divided into three departments: *Boquerón*, *Alto Paraguay* and *Presidente Hayes*. These, in turn, are divided into 15 districts (Gill et al., 2020; REDIEX, 2009).

Approximately 24.7% of this region is under a legal conservation regime as part of the National System of Wildlife Protected Areas (SINASIP)¹ (Gill et al., 2020). The Paraguayan Chaco is subdivided into five ecoregions: *Médanos*, *Cerrado*, *Pantanal*, *Humid Chaco* and *Dry Chaco* (Resolution of the Secretariat of the Environment [SEAM] N°614, 2013), each covering 3.1%, 5.1%, 17.4%, 21.5% and 52.8% of the total area, respectively (Gill et al., 2020).

The Environmental Management Directorate (DOA) classified Paraguayan Chaco's vegetation during its Chaco Environmental System

¹ The legal conservation regime is composed of protected areas, the Chaco Biosphere Reserve and indigenous communities (Gill et al., 2020). The government declares a protected area, whether public or private, in response to a request for declaration that has a technical justification containing a general diagnosis of the particular characteristics of the biological, physical and cultural resources existing in the area and their importance for the current and future conservation of ecosystems, ecological processes and natural resources (Law No. 352/1994 on Protected Wildlife Areas).

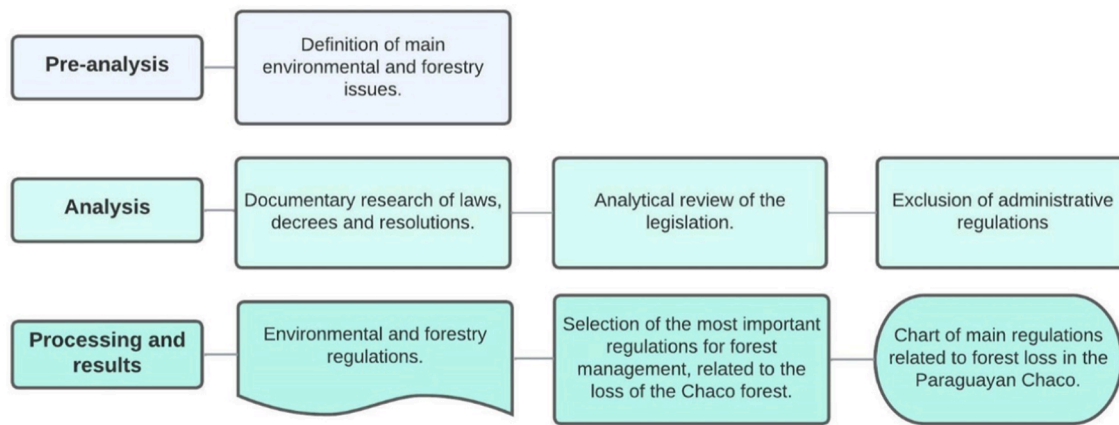


Fig. 2. Stages of the documentary analysis of the environmental and forestry legislation in force during the study period.

project in 1998 as follows: xerophytic forest, modified xerophytic forest, *Quebracho Colorado* forest, sloped forest, forest on plateaus, flooded forest (*algarrobal*, *labonal*, *palmar*, *paloblanco* and modified flooded forest), *Vallesia* and *Carob* forest, closed xerophytic forest, scrubland (scrubland on dunes, modified scrubland on dunes, open scrubland) and *Espartillar* (closed, salt marsh, aquatic vegetation and grassland) (Bartrina, 2007).

The Paraguayan Chaco has a tropical humid and a tropical dry climate that borders on a semi-arid climate. It is characterized by an average temperature of 24 °C and average annual rainfall of 600 mm near the border of Argentina and Bolivia, varying from 500 to 1,000 mm in the rest of the region, where periods of flooding alternate with very dry periods and extensive areas that are deserts in winter become waterlogged areas during the summer (FAO, 2016). The forests of the Chaco are characterized by thorny, dry shrubs, a result of high temperatures and periods of drought; the soils are classified as sandy, clayey and saline (Verijdt, 2013).

About 8.6% (20,486 km²) of the Chaco are protected areas under the International Union for Conservation of Nature (IUCN) management categories or their local equivalents, such as national parks, natural monuments, nature reserves, and managed resource reserves (Gill et al., 2020).

The Chaco Biosphere Reserve occupies 19.2% of the region (47,073 km²). Within its boundaries, six protected areas are classified as core areas: *Defensores del Chaco* National Park, *Teniente Agripino Enciso* National Park, *Médanos del Chaco* National Park, *Río Negro* National Park, *Cerro Cabrera-Timané* Natural Reserve, and *Cerro Chovoreca* Natural Monument; outside of these protected areas, there are public and private properties that are allowed to change the land use, but must maintain 50% of their surface in natural conditions in compliance with Resolution SEAM N°200/01 “By which the management categories are assigned and regulated; zoning and uses and activities”, unlike the rest of the Chaco territory where only 25% of the forest cover of a property must be maintained unaltered according to Forestry Law N°422/73 (Alonso et al., 2020; Gill et al., 2020).

The Paraguayan Chaco covers 61% of the national territory in terms of land area but presents the lowest population density. Only 3% of the total population resides in this region (Gill et al., 2020; Ortega, 2013). The first territorial organization of the Paraguayan Chaco was constituted by indigenous communities and since 1920 by Mennonite colonies that settled in the center of the region (le Polain de Waroux et al., 2017; Verijdt, 2013), but since the adoption of democratic government reforms in the early 1990 s, migration of Brazilians and more Paraguayans to the region increased (Caldas et al., 2013).

The indigenous population living in the Paraguayan Chaco represents 47.9% (54,161 inhabitants) of the country's total indigenous population (DGEEC, 2012). The 15 indigenous communities that inhabit

Chaco soil are the Ayoreo, Ybytosó, Tomaráho, Nivacle, Maka, Manjui, Angaité, Enlhet Norte, Enxet Sur, Guaná, Sanapaná, Toba Maskoy, Qom, Guaraní Occidental, Guaraní Nandeva (Paraguayan Institute of the Indigenous [INDI], 2021). Some (but not all) of the indigenous communities' lands, about 4.2% of the region (Gill et al., 2020), are included in SINASIP as a special management category of protected wilderness areas by the Resolution of the Ministry of Environment and Sustainable Development (MADES) N°562/2017.

The Mennonite colonies have approximately 18,500 inhabitants and are divided into three main areas: Menno (Loma Plata), Fernheim (Filadelfia) and Neuland. All of these are dedicated to agricultural and livestock production (Ortega, 2013).

The dominant economic activity in the region is cattle ranching, which is also the main driver of forest loss (Caldas et al., 2013; Earthsight, 2020). Cattle ranching is sustained by the productivity of natural grasslands and artificial pastures, while land for livestock production is mainly converted from dry forests to agricultural fields (Gill et al., 2020).

2.2. Case study selection

Each case study covers the term of a presidential mandate within the study period considered: (1999–2003), (2003–2008), (2008–2012), (2012–2013), (2013–2018) and (2018–2021). All of them were democratic governments (National Constitution of Paraguay, 1992), and together they represent 4 of the 25 existing political parties: The National Republican Association-Colorado Party (ANR-PC), The Patriotic Alliance for Change (APC), The Frente Guasu (FG) and The Authentic Radical Liberal Party (PLRA).

To analyze the environmental forestry regulations in force in the Paraguayan Chaco during the different periods of government, a documentary analysis technique was adopted with a qualitative approach to determine the most important environmental regulations for forest management. This technique consists of a qualitative analysis of the content through a detailed and systematized review of existing literature, reports and studies within the context of the study in order to have full knowledge of the issues related to the research (Toro-Jaramillo and Parra-Ramírez, 2010). This technique is explained in detail below (Fig. 2).

First, in the pre-analysis stage, the main issues associated with the natural forest resource were collected during the documentary research: government institutions in charge of forest management, protected wild areas, environmental protection regulations, forest management, afforestation, reforestation, and environmental impact assessment.

A database was created with the laws, decrees and resolutions of the Ministry of Environment and Sustainable Development (MADES) and National Forestry Institute (INFONA) that were enacted, repealed, and

applied, and that addressed the main issues. The complete list of laws, decrees, SEAM-CONAM/MADES resolutions and SFN/INFONA resolutions compiled and analyzed can be found in this study's appendices (Tables A2–A5).

In the next stage, a content analysis was conducted, following the methodological concepts defined by FAO (2010) for analyzing government legislation: definitions and principles, institutional regime, promotion or incentives for sustainable forest development, forest management, and land use change.

An analytical review was carried out to select the regulations that, due to their scope, the objective of the law and content related to forest management, could impact on the reduction or increase of forest loss in the Paraguayan Chaco.

To identify the central ideas of these regulations, they were classified into three groups according to their legal norms: prohibitive norms, imperative norms, and permissive norms (López Hernández, 2005).

2.3. Sources of Paraguayan Chaco forest cover data

For the analysis of annual losses, we used the data of Da Ponte et al. (2021), who estimated losses of forest and non-forest cover (including cropland, natural fields, wetlands, human settlements and other non-forest land (FAO, 2020a)) in the Paraguayan Chaco from 1987 to 2020. This dataset was created from Landsat satellite images and constitutes a dense and complete time series of the study area with validated and replicable scientific methodologies (Da Ponte et al., 2021).

To determine the initial forest cover at the beginning of each government in the years 1999, 2003, 2008, 2012, 2013, 2018 and the forest cover at the end of the year 2021, the methodology employed by Da Ponte et al. (2021) was applied, using the Random Forest algorithm through the cloud-based geomatics platform Google Earth Engine with coding in the JavaScript language.

Landsat satellite images between 1999 and 2021 were used, with cloud cover between 0 and 30% of Collection 1 - Surface Reflectance with terrain correction (L1T), so they did not require additional geometric rectification. Banding correction was applied to Landsat 7 satellite images to remove gaps and atmospheric correction was performed on all images used. The time scale used in each satellite image mosaic corresponds to one year up to the date of change of government, with the exception of 2003, 2008 and 2012, when it was extended to two years due to the inadequate quality of the images (high percentage of cloud cover) and satellites were combined as needed.

The verification of the classification results was performed considering the Overall Accuracy, User Accuracy, Producer Accuracy and Kappa Index. All forest cover classifications were reclassified with an area of two or more hectares considering the definition of forest shown in Decree N° 17.201/02 (native or autochthonous ecosystem, intervened or not, regenerated by natural succession or other forestry techniques, occupying an area of two or more hectares, characterized by the presence of mature trees of different ages, species and varied size, with one or more canopies covering more than 70% of that area and where there are more than 70 trees per hectare of 15 or more centimeters in diameter at chest height). This definition has been considered throughout the text since it seeks to associate it with local environmental and forestry legislation.

The different case studies analyzed present different time periods and cannot be directly compared with each other, so in order to obtain more information on the losses occurred during each government and to understand the overall panorama of each administration, they were analyzed from three perspectives: total forest loss in the study period (1999–2021); forest cover at the beginning of their respective mandates, considering that each government started with a different forest cover; and the average annual loss during each mandate, to detect the intensity of losses during each period.

The methodology used to determine the connection between forest cover losses in the Paraguayan Chaco and the environmental and

forestry legislation applied during each government period consisted of a qualitative analysis of the regulations, an overview of the losses that occurred during each mandate and a comparison with annual trends of forest loss before and after the enforcement of each regulation. Other factors related to forest losses such as population density, price and capacity of land use, health situation, were left out of the scope of this research.

3. Description of the six time periods considered

The presence of different political parties or presidential mandates can influence deforestation rates, preventing governments from making a real commitment to resource protection. Different ideologies, levels of corruption, the varying nature of political will, and other government priorities, can lead to different regulations on environmental protection and resource management, benefiting different sectors. Likewise, this instability affects the continuity of environmental policies put in place by each mandate (Abman, 2018).

3.1. Period of government (1999–2003)

Luis González Macchi assumed the presidency from March 28, 1999, until August 15, 2003, as the candidate of the ANR political party. He presented two reports about the Convention on Biological Diversity, which was ratified by Law No. 253/93. He also approved the Kyoto Protocol to the United Nations Framework Convention on Climate Change (Law No. 1447/99) and created the National Environmental System (SISNAM), the National Environmental Council (CONAM) and the Environmental Secretariat (SEAM) (Law No. 1561/00, Regulatory Decree No. 10579/00).

In addition, he established the National Climate Change Program (Decree No. 14,943/01) and instituted and regulated the Program for the Implementation of Environmental Measures (PIMA) (Decree No. 17726/02).

One of the most important actions of this government was the approval of the MERCOSUR Environmental Framework Agreement, in which the member states reaffirmed their commitment to the principles set forth in the 1992 Rio de Janeiro Declaration on Environment and Development (Law No. 2068/03), from which the National Biodiversity Strategy was prepared in 2003.

3.2. Period of government (2003–2008)

Nicanor Duarte Frutos was a member of the ANR political party. He assumed the presidency on August 15, 2003, and was in office for 5 years. Within a context of economic stagnation, financial and banking problems and instability in public accounts that resulted from the financial crises of its regional partners and the fall in international prices of export products, this government prioritized actions to stabilize the macroeconomic situation of the country. At that time, the economy was based mainly on agricultural production and exports, which led to an indiscriminate conversion of land for agricultural use, generating environmental and social conflicts as a result of the lack of policies and weak compliance with regulations. This was in contradiction with the country's strategic development objectives of sustainable economic growth with social, fiscal, and environmental responsibility, focused on the preservation of environmental capital (European Commission, 2007).

In 2004, this government created the National Institute for Rural and Land Development (INDERT) (Law No. 2419/04). In 2005, Duarte Frutos established the National Environmental Policy (PAN) (CONAM Resolution N°04/05), which lacked clear goals and indicators, projects or programs for the set objectives and did not include citizen participation. The following year he established the National Forestry Policy with a view to optimizing the contribution of the forestry sector to the social and economic development of the country (Gauto Ortigoza, 2019).

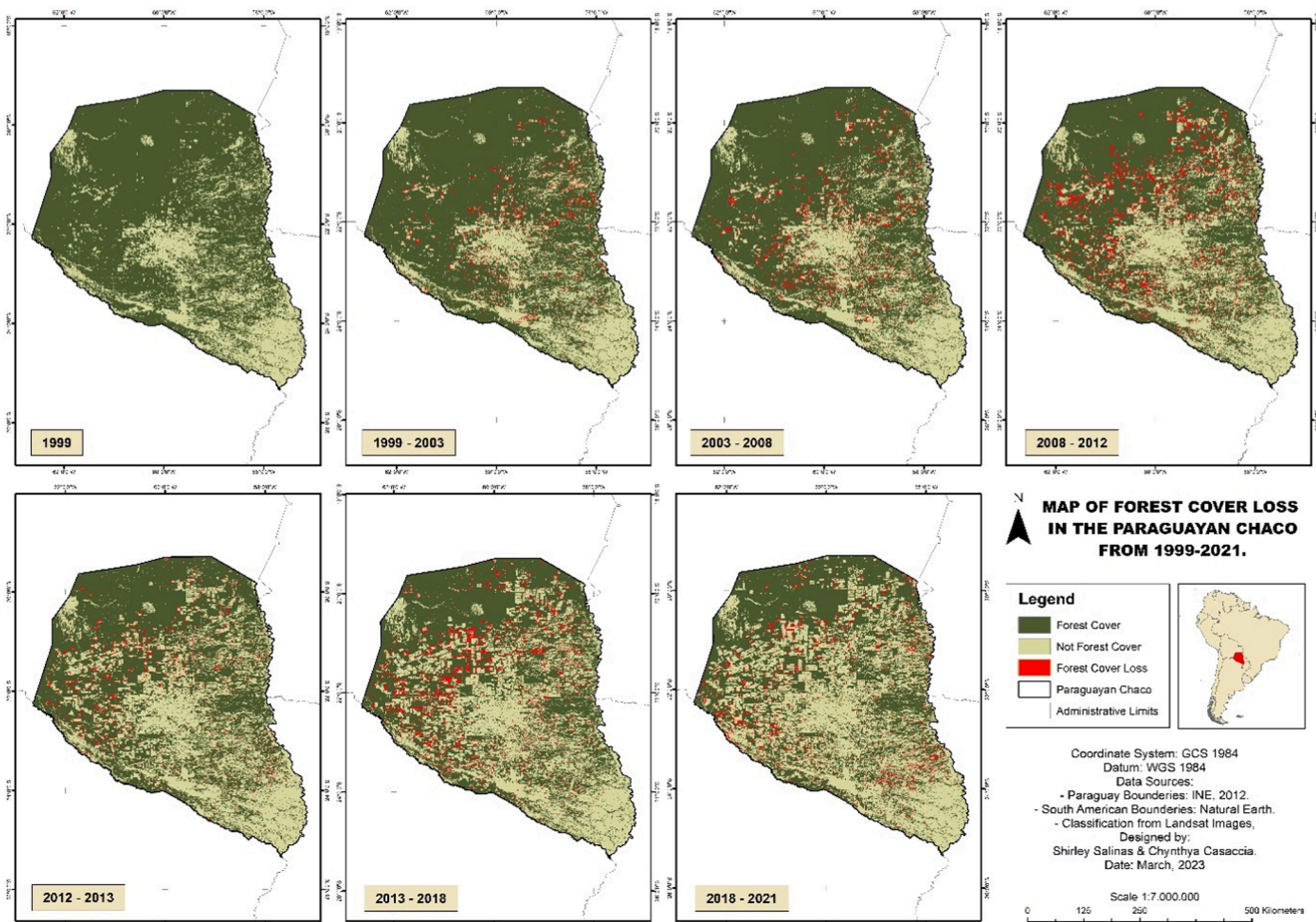


Fig. 3. Forest cover loss in the Paraguayan Chaco from 1999 to 2021.

Finally, he created the Special Registry of Properties with Natural Forests (MAG Resolution N°84/07), which featured those in the eastern region while ignoring the forests of the Chaco; and the National Forestry Institute (INFONA), which is in charge of the application of Law N°422/73 “Forestry”, Law N°536/95 “On the promotion of forestation and reforestation” and other legal norms related to the forestry sector (Law N°3464/08).

3.3. Period of government (2008–2012)

Fernando Lugo assumed the presidency on August 15, 2008, as leader of the Patriotic Alliance for Change (APC) (2008–2010) and Guasu Front (2010–2012) political parties. Representing a center-left coalition, this president was the first candidate to win the elections that did not belong to either of the two traditional parties (ANR and PLRA), after 61 uninterrupted years of the ANR in the presidency. His mandate was characterized by the implementation of meager social reforms that did not impact the established system. He governed for 3 years and 10 months until his impeachment on June 22, 2012.

This government implemented Law N°3464/08 creating INFONA, stating that SEAM is a coadjutant in the establishment of environmental criteria to be applied in forestry matters (Decree N°3929/10), and created the Department of Forestry and Environmental Affairs under the Technical Directorate of the National Police (Law N°4012/10).

Within the framework of the UNDP Regional Project “Climate Policies 2012”, Lugo prepared the National Climate Change Policy, which was approved by the National Climate Change Commission and the National Environmental Council.

3.4. Period of government (2012–2013)

Federico Franco took office as president, representing the PLRA party, on June 22, 2012, and governed until August 15, 2013. During this time, he approved a series of reforms concerning the primary sector and its exports. This government prioritized the expansion of agribusiness by intensifying the course of an extra-activist and agro-export economic model established in the nation. Franco favored the rapid growth of agribusiness with facilities and guarantees, developed an unconstitutional agrarian counter-reform, promoted mechanized agriculture through government policies and introduced biotechnological inputs in the country. Furthermore, he signed Decree N°9699/12 into law, which restructured the National Commission for Agricultural and Forestry Biosecurity (CONBIO) (Rojas Villagra, 2014).

He also declared the implementation of the “National Reforestation Plan” of national interest (Decree N°10174/12). Although this plan seeks to reduce the pressure on native forests, it was not carried out for conservation purposes but only to satisfy the demand for local biomass. He established the ecoregions for the eastern and western regions of the country (SEAM Resolution N°614/13).

3.5. Period of government (2013–2018)

Horacio Cartes assumed the presidency on August 15, 2013, representing the ANR political party for 5 years. He established the National Development Plan 2030, whose three strategic axes are poverty reduction and social development, inclusive economic growth, and the insertion of Paraguay in the world in an adequate manner (Decree N°2794/14).

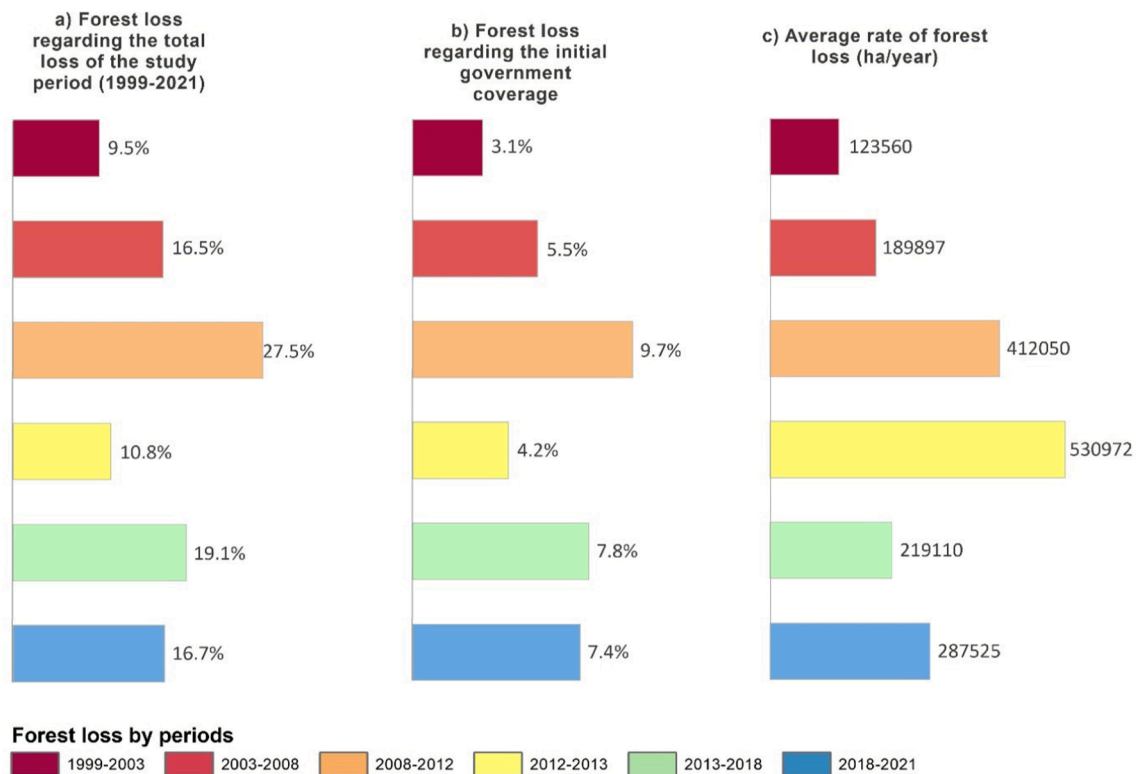


Fig. 4. Forest Cover Loss in Paraguayan Chaco by Governments.

Moreover, he declared the forestry programs and undertakings that enable and encourage the sustainable use of the soil and the promotion of biomass generation for energy and cellulose (Decree No. 2285/14) as national interests. He established regimes for the certification, control, and promotion of the use of biotechnologies (Decree No. 4056/15).

During his term the Law No. 5681/16, the “Paris Agreement on Climate Change”, and the National Law on Climate Change (Law No. 5875/17) were approved, with the purpose of establishing a general regulatory framework to plan and respond to the effects of climate change. This law led to the creation of the National Directorate of Climate Change and the National Commission on Climate Change (CNCC).

In addition, this government prepared the National Climate Change Mitigation Plan and Action Programs in 2017. Cartes elevated SEAM to the rank of Ministry, renaming it the Ministry of Environment and Sustainable Development (MADES), entrusting it with the authority to manage the PAN previously established under Duarte Frutos, governed by Law No. 1561/00 and being the authority in charge of environmental regulations (Law No. 6123/18).

One of the environmental achievements of this government was its recognition by the Green Climate Fund (GCF) as an example of environmental management for reducing deforestation and greenhouse gas emissions during the period 2015–2017, which granted over 70 million USD to the state as a reward. However, [Achucarro and Costa \(2020\)](#) state that there is no evidence linking the reduction of deforestation and GHG emissions with the implemented policies, mentioning that the funding proposal submitted by PNUMA and MADES openly stated that there were no studies on the cause of this reduction.

3.6. Period of government (2018–2021)

The current leader of the country, Mario Abdo Benítez assumed power on August 15, 2018, and is affiliated to the ANR party.

This government set environmental indicators to monitor the environmental situation and contribute to the design of environmental

strategies and policies through defined parameters, within the framework of the Capacity Development Project for the Improvement of Decision-Making Related to the Global Environment (MADES Resolution N°84/18). The environmental indicators approved were 33, which are divided into six environmental components: soil, air and climate, biodiversity, surface water and groundwater.

He passed Law N°6286/19 “On the Defense, Restoration and Promotion of Peasant Family Agriculture” into law, which has the purpose of promoting the protection of life and production systems that preserve biodiversity, as well as the sustainability, diversification, and transformation of productive systems. Additionally, he approved the National Forest Strategy for Sustainable Growth (ENBCS) (MADES Resolution N°293/19), established the classification of infractions to the Environmental Legislation (MADES Resolution N°356/19), whose enforcement authority is MADES (Decree N°1837/19) and he regulated the National Forest Monitoring System (SNMF) (Decree N°3246/20).

4. Results

4.1. Forest cover loss during the government period 1999–2021

In 1999, forests covered 73,9% (17.804.059 ha) of the Chaco territory; in 2021 the forest cover had decreased to 50,1% (12.055.890 ha), representing a loss in forest cover of 32,3% (5.748.169 ha).

In 1999 there was barely any forest cover in the center and south of the Chaco territory. Forest cover losses in 1999–2003 occurred mainly in the northeast and western center of the region, with an increasing trend in the west of the central Chaco in the following period, increasing throughout the region in the 2008–2012 period ([Fig. 3](#)). After that, a decreasing trend is observed, with continuous losses in the north and west, with a significant increase in the period 2013–2018; and again, a decrease in forest losses in the last government period (2018–2021), although they increased in the south of the region ([Fig. 3](#)).

The year 2019 was with significant forest cover losses, during the period (2018–2021), mainly affecting the *Cerro Chovoreca* Natural

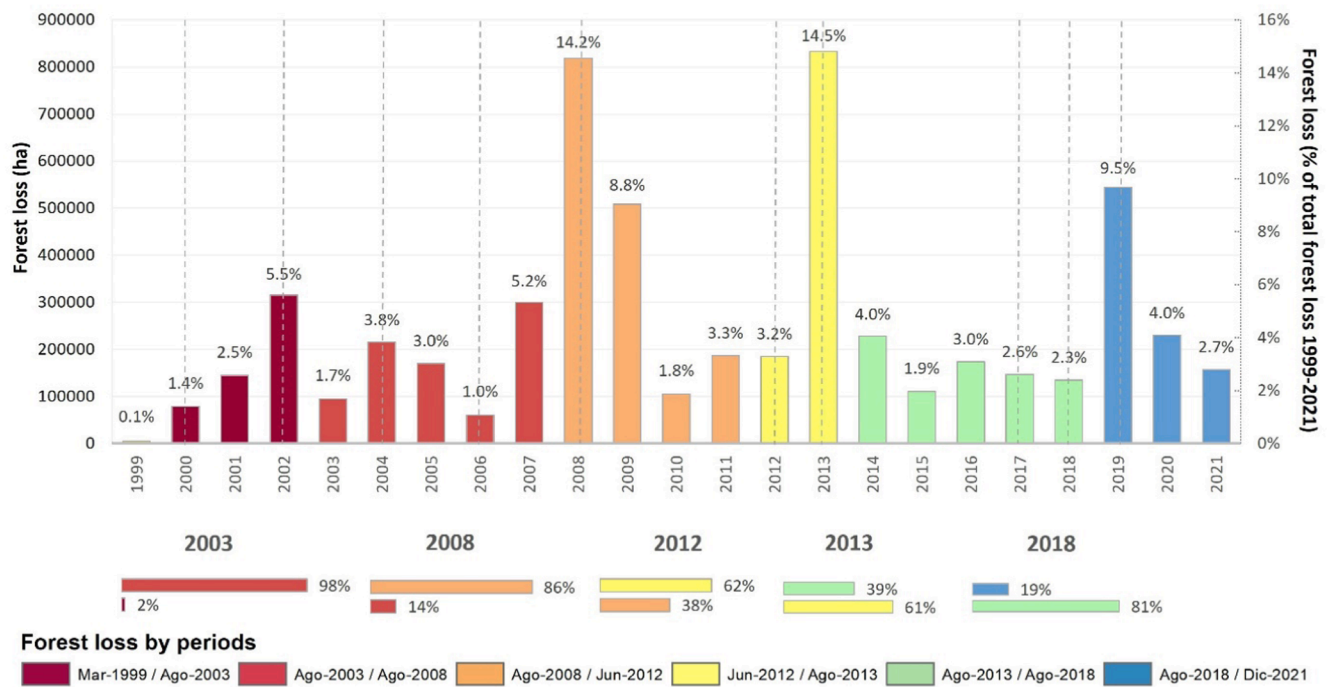


Fig. 5. Forest cover loss in the Paraguayan Chaco.

Monument located within the Chaco Biosphere Reserve. This loss of forest cover probably led to the declaration of an environmental emergency in the affected areas (Law N°6373/19, Law N°6623/20, Law N°6627/20).

Out of the six governments analyzed during the study period, the government (2008–2012) was the one that presented the greatest loss of forest cover with 27,5% (1.579.524 ha) of the total forest cover loss, followed by the periods (2013–2018) with 19,1% (1.095.552 ha), (2018–2021) with 16,7% (958.417 ha) and (2003–2008) with 16,5% (949.485 ha) (Fig. 4.a).

In contrast, the periods of governments (1999–2003) and (2012–2013) showed the lowest rates of forest loss with 9,5% (545.724 ha) and 10,8% (619.467 ha) of total forest cover loss.

It is important to highlight that forest loss regarding the initial coverage of each mandate and with regard to total losses in 22 years of study follow the same trend, but not when analyzing the average annual losses of each government (Fig. 4.a and Fig. 4.b).

When analyzing the annual trends, the period (1999–2003), stands out for having the lowest mean annual loss rates (123.560 ha/year). The following government period (2003–2008) shows an increase in the mean annual forest loss (189.897 ha/year), an increase that continued during the next governments periods (2008–2012) with 412.050 ha/year and (2012–2013), which registered a mean annual loss rate of 530.972 ha/year, the highest registered for our study timeframe. Thereafter, a decreasing trend is observed during the period (2013–2018) (219,110 ha/year), but with an increase during the following period (2018–2021) (287,525 ha/year) (Fig. 4.c).

The 2008–2012 period (Fig. 4) is characterized by immense losses that occurred regarding the government's initial forest cover and the total loss in an intensified manner due to the high average annual loss rate. An analysis of average annual forest losses by each government reveals that significant deforestation occurred in relatively short periods (2008–2012; 2012–2013; 2018–2021), in which the highest annual forest losses occurred in the years 2008, 2013 and 2019 (Fig. 5).

Upon analyzing the losses by each government, regardless of the

duration, it is observed that the highest rates correspond to administrations that completed their terms of office: (2003–2008) and (2013–2018).

4.2. Forest-related legislation during the period studied

During the period of study considered (1999–2021) a number of forest-related laws and regulations were either already in place or promulgated during this period. These laws and regulations have different implementation effects depending on their nature, as they can either forbid, order, or tolerate actions (Fig. 6). The full list of the laws, decrees, SEAM-MADES-CONAM resolutions and SFN-INFONA-MAG resolutions compiled and analyzed can be found in this study's appendices.

The importance of Article 42 of the Paraguayan Forestry Law requires especial consideration. A number of Decrees have been enacted to develop this Article (Table A1).

With the first regulation through Decree N°7,031/17, during the period (2013–2018), guidelines were established for the first time for the determination of the forest area corresponding to the legal reserve from 1986 satellite images, specifying that reforestation to cover the deficit was to be carried out for conservation purposes and with at least 40% of native species. It was specified that the Environmental Services Certificates (CSA) must correspond to the same ecoregion where the deficit is located and that the habilitations (irrational clearing or felling) within legal reserves would not fall under the regime of payment for environmental services. It was ruled that in this case the compensation would be made with reforestation of native species in its entirety and, without exception, the payment of the corresponding fine would not be waived.

Despite improvements introduced with this decree, it was promptly repealed months later by Decree No. 7674/17. The acquisition of CSAs or reforestation with native species was approved, in addition to the payment of the corresponding fines for those who carry out clearances in legal reserves. Effectively, this constituted a license to log and pay, with

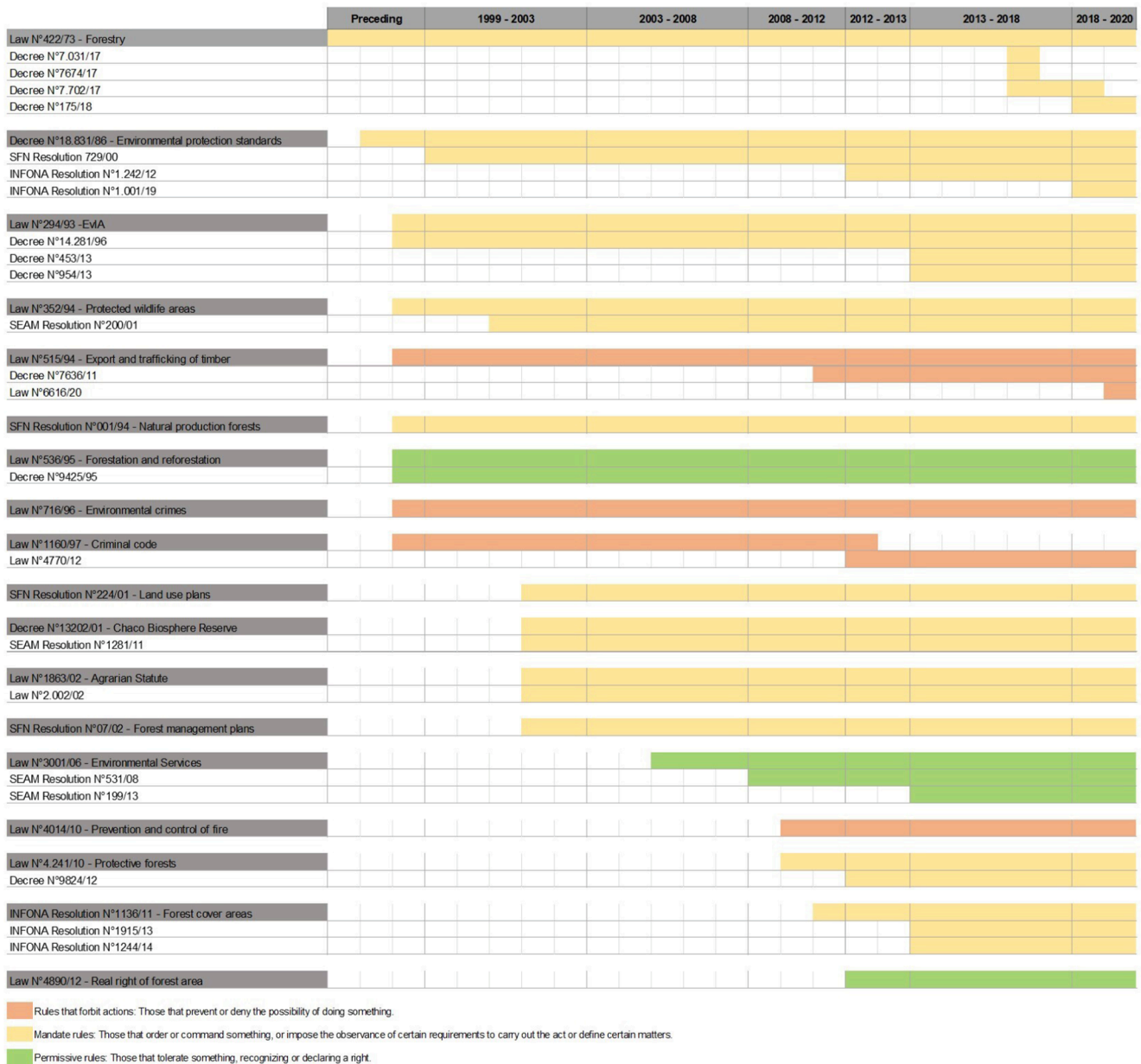


Fig. 6. Most relevant forest-related regulations applied to the Chaco region during the study period.

the profits generated by illegal deforestation being higher than the sum of the cost of the certificates plus the INFONA sanctions (WWF, 2017a), including the possibility of buying CSAs in advance (without necessarily being from the same ecoregion where the deficit is located) and having “permission to deforest in the future”. However, this decree could not have caused negative effects as it was repealed 10 days later by Decree N°7. 702/17.

The latter represented a major setback aggravating the points mentioned above, by allowing reforestations to cover allotments within the reserves to be carried out with an agroforestry system and without specifying that they must be carried out for conservation purposes, “opening the margin of excessive interpretation of certain aspects of legislative technique that allow its inappropriate use” as mentioned by INFONA president Cristina Goralewski (ABC Color, 2018). Finally, it was repealed by Decree N°175/18 during the current government (2018–2021), which reestablished Decree N°7031/17.

5. Discussion

The main environmental regulation in Paraguay is Forestry Law No. 422/73, which in Article No. 42 establishes the obligation to a) legally maintain, 25% as a reserve in rural lands of more than 20 ha within forest areas, and b) in the case of not having this percentage, to reforest up to 5% of the area. The second indirectly enables landowners to legally deforest 75% of their forests. None of the governments have modified this regulation and it is the one that favors the loss of forest cover the most, not even being complied with on several occasions (Carrere, 2019; IDEA, 2020; Manzoni et al., 2019).

This study highlights the existence of legal loopholes in the legislation since there is no law prohibiting the transfer of ownership of the area where the forest reserve is located, which can be cleared by the new owner, thus reducing the area of the existing natural forest.

One of the regulations that have most influenced forest loss is the Agrarian Statute of 1963, which generated the belief that forest lands are unproductive and subject to expropriation, having more value as pasture

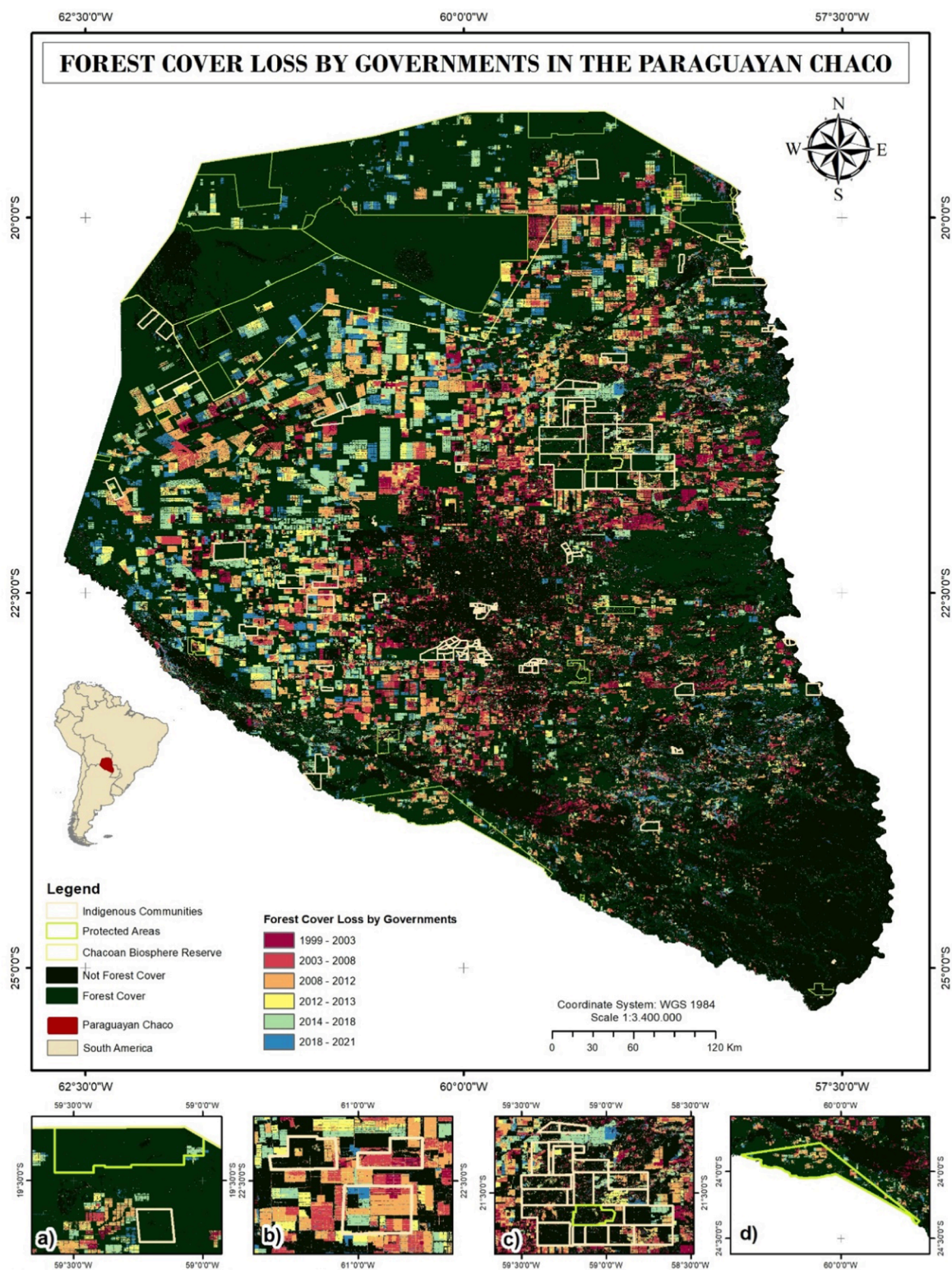


Fig. A1. Spatial distribution of forest cover loss by government periods in the Paraguayan Chaco.

or cropland, thus justifying their use; implying an incentive for deforestation in a speculative manner (Kernan et al., 2010; Veit and Sarsfield, 2017). This law was modified by Law No. 2,002 in 2002, during the period (1999–2003). It is noteworthy that the government had the

lowest forest cover losses in the 22 years included in this analysis, and that it legally vindicated the productive and environmental value of forests.

The government (2003–2008) had the second lowest average annual

Table A1
Regulatory Decrees of Article 42 of the Forestry Law.

Important Aspects	Decree N°7031/17	Decree N°7674/17	Decree N°7702/17	Decree N°175/18
In case of a deficit of the legal reserve stipulated in the Forestry Law (when acquiring the property)	CSAs can be acquired, or reforestation can be done with native species in their entirety or by applying mixed forest plantation*	CSAs can be acquired, or reforestation can be done with native species in their entirety or by applying mixed forest plantation*	CSAs can be purchased, or reforestation can be done with native species in their entirety or by applying mixed forest plantation*	CSAs can be acquired, or reforestation can be done with native species in their entirety or by applying mixed forest plantation*
In the event of dismantling the existing legal reserve on the property	CSAs cannot be acquired, and reforestation cannot be by mixed forest plantation, and must be entirely with native species	CSAs can be acquired, or reforestation can be done, but without using mixed forest plantation, and must be entirely with native species	CSAs can be acquired, or reforestation can be done, allowing mixed forest plantation or with native species in its entirety	CSAs cannot be acquired, and reforestation cannot be by mixed forest plantation, and must be entirely with native species
Ecoregion where the CSA are located	Same ecoregion	Indistinct	Indistinct	Same ecoregion
Type of CSA to be acquired	Not specified	Not specified	CSA of natural forests	Not specified
Restoration for conservation purposes	Yes	Yes	Not specified, thus allowing the possibility of use of the reserve	Yes

CSA: Environmental Services Certificates.

* The agroforestry system with mixed forest plantation characteristics may not have a percentage of less than 40% of native species.

forest loss rate. It stands out for the enactment of two important laws. The first one is the Zero Deforestation Law in the Eastern Region of the country, extended by all the following mandates (Law N°3.139/06, Law N°3.663/08, Law N°5.045/13, Law N°6.256/18, Law N°6.676/20), which resulted in an increase in the pressure on Chaco's forests and can also be evidenced with the significant increase in forest losses since 2007.

The second one is Law No. 3,001/06 "On the valuation and retribution of environmental services". However, this law lacks a regulation for its different modalities, with the risk of setting a fee to an invaluable resource, not taking into account that forests have different types of value, opening the possibility that those who have not met the requirement of legal reserve can access the Certificates of Environmental Services (CSA, by its Spanish acronym) to cover the deficit without the need for reforestation, reaching at best the conservation but not the increase of the legal reserve capital of the forests; since its implementation has not been perceived any decrease in deforestation in the Paraguayan Chaco.

During the period (2008–2012), much of the losses were caused by members of the Paraguayan Rural Association, mainly Mennonites, who stated that they were cutting down forests for fear of the increasing restrictions on the number of forests to be cleared and being trapped on a land that does not generate profits for them, mentioning that they accept that ecological damage is caused by their actions (Vidal, 2010). In 2009, despite continued forest losses, the Zero Deforestation Bill for the Chaco was rejected due to opposition from the agrarian union (le Polain de Waroux et al., 2016; le Polain de Waroux et al., 2019).

Highlights during this term in office include regulations that favor forest conservation, such as Law No. 4,241/10 for the re-establishment of forests protecting watercourses and Law No. 4011/10 for fire prevention and control, although this law is inapplicable because the municipalities do not have the necessary tools to control where the fire is happening, the technical capacity or the resources to enforce the law; its enforcement should be a joint effort from INFONA, MADES, the National Meteorological Center and the Prosecutor's Office (La Nación, 2021).

It can be inferred that this government's losses were not related to the enactment of specific regulations or the repeal of existing ones, but rather the lack of monitoring and compliance with existing regulations.

During the period (2012–2013), no regulations were enacted that were directly related to forest loss. However, deforestation during this government is mainly related to the expansion of the livestock frontier (FAO, 2013), since a million-dollar credit line for cattle ranching was made available during this administration (Góngora Sánchez, 2021).

The following period (2013–2018) has the second highest forest losses in the 22 years studied. It is worth noting that in the first months of this administration, the Decree N°14281/96 regulating Law N°294/

93 was repealed, which was considered less permissive than its amendments (Decree N°453/13, Decree N°954/13). These decrees created a detailed list of all activities that require EvIA. They are permissive for certain activities and can become extremely harmful to the environment due to their numerous exemptions, such as the cases of agricultural or livestock exploitation in properties smaller than 2000 ha in the western region, reforestations or forestations in areas smaller than 1000 ha or 500 ha in the eastern region, land clearing or land use changes with natural forests smaller than two hectares, among others. In addition, these regulatory norms violated the basic principle of priority of laws. They established measures not originating in the law, increasing the lack of protection of the environment and the population (Rojas Villagra, 2014).

These decrees are considered to probably have contributed to the high rate of forest loss during the period (2013–2018), but their impact could not be determined with precision. For this, an analysis focused on the surface areas of the different properties located in the Chaco is required, a level of detail not foreseen in this research.

During this mandate, in 2017, Art. 42 of the Forestry Law was regulated on three occasions (Decrees N°7031/17, N°7674/17, N°7702/17), making it, on each occasion, more permissive and more open to interpretation. With Decree N°7702/17, the possibility of exploiting the native forest to the point of leaving it totally degraded was opened, constituting a transgression to the principle of non-regression in environmental matters established in the Rio + 20 International Conference. This decree allowed the increase of legal deforestation related to livestock expansion within the Chaco. While it was in force many farms completely cleared their land (Benítez, 2018).

The reforestation plans implemented by this government (Decree No. 2285/14; Decree No. 3050/15), were carried out for energy and commercial purposes, not for landscape restoration; they were intended to cover the deficit of forest biomass for energy generation. It should be noted that, this mandate stated that its objective was to reduce greenhouse gas emissions, yet the decrees enacted during this period went against those ideals, which is why professionals from different areas, environmental organizations, and civilians in general expressed their concern for the established regulations (González Vera, 2013; Monges, 2017; WWF, 2017b).

The third highest rate of average annual forest loss took place under the current government (2018–2021). It stands out for repealing Decree N°7702/17 through Decree N°175/18, and temporarily reinstating Decree N°7031/17 until the presentation of a new regulation. It is worth noting that there is currently no new regulation; however, with the reinstatement of Decree N°7031/17, a decrease in the loss of forest cover was observed in 2018, but not in 2019, a year heavily punished with forest fires in the Paraguayan Chaco.

Table A2
Complete list of the Laws analyzed.

Law	Name
Law N°854/63	Establishes the Agrarian Statute.
Law N°422/73	Forestry Law.
Law N°11.681/75	Regulating Forestry Law No. 422.
Law N°43/89	Modifying provisions of Law No. 1372/88 "Establishing a regime for the regularization of settlements of indigenous communities".
Law N°40/90	Creates the National Commission for the Defense of Natural Resources.
Law N°81/92	Organizational and functional structure of the Ministry of Agriculture and Livestock.
Law N°96/92	Of wildlife.
Law N°251/93	Approving the Convention on Climate Change adopted during the United Nations Conference on Environment and Development - The Earth Summit - held in the city of Rio de Janeiro, Brazil.
Law N°253/93	Approving the "Convention on Biological Diversity", adopted during the United Nations Conference on Environment and Development -The Earth Summit-, held in Rio de Janeiro, Brazil.
Law N°294/93	Environmental Impact Assessment.
Law N°345/94	Amends Law No. 294/93, on Environmental Impact Assessment.
Law N°352/94	Protected Wildlife Areas.
Law N°515/94	Prohibiting the export and trafficking of logs, logs, and beams of wood.
Law N°536/95	Promotion of afforestation and reforestation.
Law N°542/95	Of forest resources.
Law N°716/96	That punishes crimes against the environment.
Law N°816/96	Adopting measures for the Defense of Natural Resources.
Law N°1.160/97	Penal Code.
Law N°1.447/99	Approving the Kyoto Protocol to the United Nations Framework Convention on Climate Change.
Law N°1.561/00	Which creates the National Environmental System, the National Environmental Council and the Environmental Secretariat.
Law N°1.639/00	Amending and extending Law No. 536 of January 16, 1995 "On the promotion of forestation and reforestation".
Law N°1.863/02	Establishment of the Agrarian Statute.
Law N°1.968/02	Amending Law No. 1639 "Amending and extending Law No. 536 of January 16, 1995 "On the promotion of forestation and reforestation".
Law N°2.002/02	Amends several articles of Law No. 1863 of January 30, 2002, which establishes the Agrarian Statute.
Law N°2.068/03	Approving the MERCOSUR Framework Agreement on the Environment.
Law N°2.419/04	Creating the Institute for Rural and Land Development (INDERT).
Law N°2.524/04	Prohibiting in the Eastern Region the activities of transformation and conversion of forest cover areas
Law N°3.001/06	Valuation and Remuneration of Environmental Services.
Law N°3.139/06	Extending the validity of Articles 2 and 3 and extending Law 2524/04 "Prohibiting in the Eastern Region the activities of transformation and conversion of forest cover areas".
Law N°3.239/07	Water resources of Paraguay.
Law N°3.464/08	That creates the National Forestry Institute - INFONA.
Law N°3.663/08	That amends Articles 2 and 3 of Law No. 2,524/04 "On the prohibition in the eastern region of the activities of transformation and conversion of areas with forest cover", as amended by Law No. 3,139/06.
Law N°4.012/10	Which creates the Department of Forestry and Environmental Affairs under the Technical Directorate of the National Police and specifies the functions of the National Police in environmental matters.
Law N°4.014/10	Fire prevention and control.
Law N°4.241/10	On the Reestablishment of Protective Forests of Watercourses in the National Territory.
Law N°4.770/12	Amending Article 2020 of Law No. 1160/97 "Penal Code".
Law N°4.890/13	Real Right of Forest Area.
Law N°5.045/13	Amending Articles 2 and 3 of Law No. 2524/04, as amended by Law No. 3139/06 and extended by Law No. 3663/08.
Law N°5.392/15	Which establishes the boundaries of the <i>Defensores del Chaco</i> National Park, created by Executive Decree No. 16,806 of August 6, 1975.
Law N°5.540/15	That amends Article 4° of Law N°5392/15 "That establishes the boundaries of the <i>Defensores del Chaco</i> National Park, created by Executive Decree N°16,806 of August 6, 1975".
Law N°5.875/17	National Climate Change.
Law N°6.123/18	Elevates the Secretariat of the Environment to the rank of Ministry and renames it Ministry of the Environment and Sustainable Development.
Law N°6.256/18	Prohibiting the transformation and conversion activities of areas with forest cover in the Eastern Region.
Law N°6.286/19	For the defense, restoration, and promotion of family farming.
Law N°6.373/19	Declaring the Departments of Alto Paraguay and Boquerón of the Western Region affected by forest fires in an Environmental Emergency situation.
Law N°6.616/20	Modifying and expanding Article 1° of Law 515/94 "Prohibiting the export and traffic of logs, logs and beams of wood".
Law N°6.623/20	Declaring the Department of Boquerón of the Western Region, affected by drought, forest fires and associated ecosystems, to be in a situation of Environmental Emergency.
Law N°6.627/20	That declares the entire territory of the Republic to be in a situation of National Emergency, as a consequence of the fires and the damage caused to the ecosystems and the environment.
Law N°6.676/20	Prohibiting transformation and conversion activities of areas with forest cover in the Eastern Region.

From the above analysis, it can be deduced that the increase in deforestation has been influenced by several factors, such as the lack of updated forest legislation; the low percentage of forest reserves, which are mandatory and only required for large plots; the low institutional capacity to implement and monitor the laws in an efficient way; the concentration of conservation laws centered in the eastern region of the country; and the lack of a legal framework for forest conservation associated with the sustainable development of ancestral and indigenous communities still present in the territory.

Therefore, forest losses have intensified due to the lack of state resources to implement and monitor laws and to prosecute violators (Génez Báez, 2020, Verijdt, 2013, Vidal; 2010). With high levels of corruption hindering the enforcement of regulations, deforestation is

propitiated throughout the country (Earthsight, 2020; le Polain de Waroux et al., 2016, Wright et al., 2007), along with the limited technical capacity of government institutions to perform satellite monitoring of forests and early detection of forest cover loss (le Polain de Waroux et al., 2016, Veit and Sarsfield, 2017).

In addition, a problem at the national level is the institutional absence in the field of the authorities responsible for enforcing environmental regulations due to overlapping roles, lack of human resources and tools (Génez Báez, 2020). There is no reliable database that the institutions of the sector, nor is there a rural land registry at country level that makes it possible to determine whether the properties comply with the percentage of legal forest reserve required by law, not being able to determine which deforestation is illegal or those responsible for

Table A3
Complete list of Decrees analyzed.

Decree	Name
Decree N°18.831/86	Whereby environmental protection standards are established.
Decree N°9.425/95	Regulating Law No. 536/95 "On Promotion of Afforestation and Reforestation".
Decree N°14.281/96	Regulating Law No. 294/93 on Environmental Impact Assessment.
Decree N°10.579/00	Regulating Law No. 1561/00, "which creates the National Environmental System, the National Environmental Council and the Environmental Secretariat".
Decree N°13.202/01	Creates the Chaco Biosphere Reserve, located in the Department of Alto Paraguay and the Department of Boquerón.
Decree N°14.943/01	Whereby the National Climate Change Program is implemented.
Decree N°17.201/02	By which Articles 12, subsection "n" and 15, subsection "b" of Law 1561/00 "Creating the National Environmental System, the National Environmental Council and the Environmental Secretariat" are regulated.
Decree N°17.726/02	Whereby the Program for the Implementation of Environmental Measures (PIMA) is established and regulated.
Decree N°1.791/04	By which the effects of Executive Decree No. 21,957 of August 12, 2003, declaring an area of 514,233 ha, located in the District of <i>Mariscal Estigarribia</i> , Department of <i>Boquerón</i> , under the denomination of <i>Médanos del Chaco</i> National Park, are suspended.
Decree N°2.726/04	Whereby the provisions of Article 1 of Decree No. 1,791 of February 17, 2004, and Article 1 of Decree No. 21,957 of August 12, 2003, are partially amended.
Decree N°3.929/10	Regulating Law No. 3.464/08 "Creating the National Forestry Institute (INFONA).
Decree N°7.636/11	Regulating Article 26 of Law No. 422/73 and establishing a special regime for the harvesting, transportation, and commercialization of timber and other products from forest plantations established with exotic or introduced fast-growing species.
Decree N°9.699/12	Whereby the National Commission on Agricultural and Forestry Biosecurity (CONBIO) is created.
Decree N°9.824/12	Regulating Law No. 4241/10 "On the reestablishment of forests protecting watercourses within the national territory".
Decree N°10.174/12	By which the implementation of the "National Reforestation Plan", presented by INFONA, is declared of national interest.
Decree N°453/13	Whereby Law No. 294/1993 "On Environmental Impact Assessment" and its amendment, Law No. 345/1994, are regulated and Decree No. 14,281/96 is repealed.
Decree N°954/13	Whereby Articles 2, 3, 5, 6.e, 9, 10, 14 and the annex of Decree 453/13 and its amendment Law No. 345/94 are amended and expanded, and Decree No. 14281/96 is repealed.
Decree N°2.285/14	By which Forestry Programs and Enterprises that enable and encourage the sustainable use of the soil and the promotion of Biomass generation for energy and cellulose are declared of national interest.
Decree N°2.794/14	Approving the National Development Plan (Plan 2030).
Decree N°3.050/15	Whereby the implementation of programs within the framework of the National Reforestation Plan is provided for, and Mr. Oscar Vicente Scavone Rivas, Economic and Commercial Advisor to the President of the Republic, is entrusted and empowered to coordinate said programs.
Decree N°4.056/15	Whereby the Vice-Ministry of Mines and Energy, under the MOPC, in coordination with INFONA, SEAM and the Ministry of Industry and Commerce, is authorized to establish certification, control and promotion regimes for the use of bioenergy that guarantee the sustainability of these renewable energy resources.
Decree N°7.031/17	Regulating Article 42 of Law No. 422/1973 "Forestry".
Decree N°7.152/17	Whereby Articles 3° and 6° of Decree N°7031, dated April 17, 2017, "Whereby Article 42 of Law N°422/1973 "Forestry" is regulated" are amended.
Decree N°7.674/17	Whereby Article 42 of Law N°422/1973 "Forestry" is regulated and Decrees N°7031, dated April 17, 2017, and N°7152, dated May 22, 2017, are abrogated.
Decree N°7.702/17	Regulating Article 42 of Law 422/1973 and abrogating Decrees N°7031 of April 17, 7152 of May 2017 and N°7674 of September 4, 2017.
Decree N°175/18	Whereby Decree 7702 of September 14, 2017, is repealed, the President of the National Forestry Institute (INFONA) is instructed to propose a new regulation, and an interim regime regulating Article 42 of Law 422/1973 "Forestry" is established.
Decree N°1.837/19	Whereby Article 5° of Law N°5146/2014 is regulated.
Decree N°3.246/20	By which the National Forest Monitoring System of Paraguay (SNMF) is regulated.

it (Carrere, 2019).

Within the governments studied, the losses can be considered the result of poorly focused public policies, the flexibility of permissive regulations that effectively incentives to deforest, the legal loopholes they present, the existence of few administrative and criminal sanctions, the difficulty to monitor compliance with regulations, and the lack of consideration of the regeneration capacity of forests as well as of their ecological and environmental importance.

The above results represent a wake-up call for future governments since the enactment of laws and the establishment of policies has not resulted in the reduction of forest loss. It is necessary to establish systematic monitoring and evaluation of policies, including concrete actions, designate responsible professionals, technical and economic instruments; complemented with forest monitoring and the inclusion of

public policies that take into account not only economic growth, but also equitable social distribution and the sustainability of forest resources, incorporating the use of multiple criteria with strategies that integrate all social sectors and do not benefit only one economic sector. This way, it will be possible to facilitate decision-making in the forestry sector, keep national forest programs effective and updated, and measure the level of compliance with international commitments (National Forest Commission CONAFOR, 2011; FAO, 2014).

6. Conclusions and recommendations

Average annual forest loss rates are mainly affected by the time each government spends in office, being the highest two from the governments with the shortest duration, (2008–2012) and (2012–2013), which

Table A4

Complete list of SEAM-MADES-CONAM Resolutions analyzed.

SEAM-MADES-CONAM Resolution	Name
SEAM Resolution N°200/01	By which management categories, zoning, uses, and activities are assigned and regulated.
Resolution SEAM N°531/08	By which the conditions and requirements to certify the environmental services produced by forests are established, as well as the conditions and requirements for the purchasers of forest environmental services certificates to use them to compensate the deficit of the legal reserve of natural forests, in accordance with Laws 422/73 and 3001/06.
SEAM Resolution N°1.281/11	By which the area of the Paraguayan Chaco Biosphere Reserve, created by Decree No. 13,202 dated May 27, 2001, is expanded to include the total area approved by the UNESCO MaB Committee.
SEAM Resolution N°199/13	By which the conditions and requirements to certify the environmental services produced by forests are established, as well as the conditions and requirements for the purchasers of forest environmental services certificates to use them to compensate the deficit of the legal reserve of natural forests, in accordance with Laws 422/73 and 3001/06.
SEAM Resolution N°614/13	Establishing the ecoregions for the eastern and western regions of Paraguay.
SEAM Resolution N°562/17	By which Resolution N°200/01 dated August 21, 2001 “By which management categories, zoning, uses and activities are assigned and regulated” is modified and expanded.
MADES Resolution N°84/18	Whereby the Environmental Indicators are approved.
MADES Resolution N°293/19	Approving the National Forest Strategy for Sustainable Growth (ENBCS).
MADES Resolution N°356/19	By which the regulation of typifications of infractions to the environmental legislation is established.
CONAM Resolution N°04/05	Whereby the National Environmental Policy of Paraguay (PAN) is approved.

Table A5

SFN-INFONA-MAG Resolutions analyzed.

SFN-INFONA-MAG Resolution	Name
SFN Resolution N°76/92	Replacing Resolution No. 61/92 and regulating the preparation of forest harvesting and management plans.
SFN Resolution N°001/94	Establishing norms for the protection of natural production forests.
SFN Resolution N°323/94	Modifies Resolution No. 76/92, on the preparation of forest harvesting and management plans.
SFN Resolution N°729/00	Regulating environmental protection standards in the Western Region or Chaco.
SFN Resolution N°224/01	By which the elaboration and presentation of land use plans is regulated.
SFN Resolution N°07/02	Regulating the preparation and presentation of forest management plans.
INFONA Resolution N°1.544/11	By which the registry of properties whose harvesting area is up to fifty (50) hectares of forest in the Western Region is created and the procedure for the approval of forestry work plans is established.
INFONA Resolution N°1.136/11	Establishing the reference parameters for the annual allotment of areas with forest cover in the Western Region.
INFONA Resolution N°1.242/12	By which Articles 6 and 11 of Decree No. 18.831/86 “Whereby environmental protection regulations are established” are regulated for the Western Region.
INFONA Resolution N°1.915/13	By which reference parameters and conditions are established for the annual authorization of areas with forest cover in the Western Region.
INFONA Resolution N°1.244/14	Whereby amending Article 1° of Resolution No. 1915/13, dated December 16, 2013.
INFONA Resolution N°1.001/19	Whereby Article 1° of Resolution No. 1,242/2012 dated October 16, 2012 “Whereby it regulates for the Western Region, Articles 6° and 11° of Decree No. 18,831 “Whereby environmental protection rules are established” and provides for regulation of forest strips for plots of land of less than 51 ha” is modified.
MAG Resolution N°84/07	By which the Special Registry of Properties with Natural Forests of the Eastern Region is created within the scope of the National Forest Service (SFN) and calls for registration.

reflects that in short periods of time, large deforestations occurred in the Paraguayan Chaco. It is also necessary to consider the sum of the annual losses of each government without averaging them, resulting in the highest forest loss during the government period (2008–2012), followed directly by the periods (2013–2018) and (2018–2021).

Although these governments did not enact laws that directly promoted deforestation, it is inferred that forest losses resulted from the agro-export economic model already established in the country several decades earlier.

It is important to note that forest cover losses have occurred not only in private properties but also in protected wild areas and mainly in indigenous communities, despite existing regulations. Therefore, these results show a close relationship between the loss of forest cover and the lack of compliance with existing regulations, many of which are flexible, permissive, focused on the economic and productive value of forests, and allow for subjective interpretations that are taken as incentives for the legal deforestation of forests.

This means that the mere existence of environmental public policies does not ensure their effectiveness, nor does it generate a positive impact on forest conservation if they are not correctly applied. Current and future leaders should consider establishing a multidisciplinary team to evaluate and rethink environmental public policies with the objective of improving their design by learning from the mistakes made.

To assess the effect of environmental policies on forest cover loss in greater depth, a number of factors should be considered, such as the relationship between environmental laws and the country's economic development policies, land-use plans, and the cultural environment affected, which synergistically affect land-use change. The influence of legislation on forest loss could be more accurately assessed if the level of enforcement and strictness of compliance with individual laws were considered in a more comprehensive and quantitative analysis.

It is recommended that future studies take into account the aspects detailed in the discussion and, furthermore, to:

- Evaluate whether legal regulations on agriculture and livestock had any influence on deforestation using a data set that differentiates between natural forests and forestation and that represents forest cover up to the date of change of government in order to have more accurate information, especially for governments of shorter duration.
- Determine how the production and export of charcoal affect the native forests of the Chaco, particularly in the indigenous territories.
- Analyze the loss of forest cover within each indigenous community, mainly in the territories of the Ayoreo Totobiegosode Natural and Cultural Heritage (PNCAT), which is home to the last native peoples living in voluntary isolation.

- Determine whether losses of forest cover in protected wildlife areas are related to their ownership (public or private) or to their management category and identify whether they have updated management plans.
- Evaluate the potential of local governments in forest protection based on the role of municipalities.
- Consider the interactions of forest cover loss with other factors that were not analyzed in this study, such as biogeographic location, forest fires, land use changes, quantity and quality of public spending on environmental enforcement, and loss of forests as leakage due to stricter environmental regulations in neighboring countries.

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A

(See Fig. A1).

(See Tables A1–A5).

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